Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/06/2015
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Project Title:	Defining the Role of miR-182 in HZE-Induced Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Important opportunities for space travel in the decades ahead include a proposed mission to Mars. However, an important deterrent to such interplanetary travel is the risk of radiation-induced carcinogenesis. There are many uncertainties in predicting the risk of space travel because the mechanisms of carcinogenesis from space radiation, which is dominated by high-energy protons and high (H) charge (Z) and energy (E) (HZE) particles, are poorly understood. Here, we will test the hypothesis that microRNAs that regulate the expression of proteins that are known to regulate radiation-induced cancer, will function as non-protein encoding genetic factors that mediate radiation carcinogenesis from HZE particles. To test this hypothesis, we will utilize novel genetically engineered mice in which we can delete or overexpress miR-182. Mice will be exposed to low LET (linear energy transfer) and HZE particle radiation. Based on our preliminary experiments, we anticipate that mice lacking miR-182 will be protected from radiation-induced cancer, while mice overexpressing miR-182 will be sensitized to radiation carcinogenesis. Therefore,		

	we expect that this research will establish that miRNAs are nonprotein encoding genetic factors that regulate radiation carcinogenesis. Furthermore, by studying how radiation quality impacts radiation-induced cancer in the setting of miR-182 deletion and overexpression, we will gain new insights into the mechanisms of the relative biological effectiveness (RBE)/LET relationship for carcinogenesis. Ultimately, this information can be used to define better risk estimates of cancer from deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cancer is a major cause of morbidity and mortality. Cancer arises from a combination of genetic mutations in tumor cells and alterations in the microenvironment. Further research into the mechanism of cancer development is needed so that more successful strategies for prevention and treatment of cancer can be developed. By studying the mechanisms of cancer initiation and development, we will generate new knowledge that can be used to develop novel approaches for the prevention and treatment of cancer. Here, we will study the role of miRNA 182 in radiation-induced cancer. Therefore, this work will help define the role of miRNAs in radiation-induced cancer.
Task Progress:	In our second year of funding, we performed the experiments proposed in our application by breeding and irradiating mice with X-rays at Duke University and 56Fe and 28Si particles at Brookhaven National Laboratory (BNL). In addition, members of our lab interacted with outside experts in space radiation and radiation-induced tumorigenesis by presenting at the NASA Human Research Program Investigators' Workshop in Galveston, Texas from January 13-15, 2015. Through these activities and from our research projects, we have gained new information about the effects of space radiation on normal lung tissue and lung cancer development. Project Title: Defining the role of miR-182 in HZE-induced tumorigenesis. David Kirsch, M.D., Ph.D., Lead We are testing the hypothesis that microRNAs that control the expression of proteins known to regulate radiation-induced cancer will function as genetic factors in radiation carcinogenesis. In order to study the role of microRNA-182 (miR-182) in radiation-induced lymphomagenesis and lung tumorigenesis following exposure to X-ray or space irradiation, we developed a mouse model that lacks miR-182 expression (Aim 1) or expresses an elevated level of miR-182 (Aim 2). For Aim 1, we crossed our mouse model of lung cancer (RbFL/FL; p53FL/+) to mice that lack miR-182 in the whole animal or conditionally in tumor cells. At the end of our first year of funding, we will have infected and irradiated approximately 575 of these mice with space radiation. These mice will be followed for 300 days following radiation exposure, at which point we will collect the lungs to quantify lung tumor burden. We have also completed a pilot study on miR-182 knockout animals to investigate the role of miR-182 on radiation-induced lymphomagenesis. Less than 20% of the mice developed lymphoma within 300 days and there was no significant impact of miR-182 expression level on lymphoma-free survival. Therefore, in the third year of funding, we plan to repeat these studies with a weekly fractionation regimen in order to increase the incidence of lymphoma. For Aim 2, we crossed our mouse model of lung cancer to mice that overexpress miR-182 in the whole animal or conditionally in tumor cells. At the end of our first year of funding, we will have infected and irradiated approximately 425 of these mice with space radiation. These mice will be followed for 300 days following radiation exposure, at which point we will collect the lungs to quantify lung tumor burden. Additionally, in the third year of funding, we will begin a radiation-induced lymphomagenesis study using miR-182 overexpressor mice.
Bibliography Type:	Description: (Last Updated: 11/19/2025)
Abstracts for Journals and Proceedings	Castle KD, Lee CL, Sachdeva M, Moding EJ, Luo L, Kirsch DG. "Defining the role of miR-182 in tumorigenesis after exposure to space radiation." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 13-15, 2015. , Jan-2015